P1 Particles and Radiation – Knowledge Organiser

AQA A-level Physics • Set 1: Constituents of the Atom **Paper 1**

#	Question	Answer
1	Charge of a proton	$+1.60 \times 10^{-19}$ C
2	Charge of an electron	-1.60×10^{-19} C
3	Mass of a proton	1.67×10^{-27} kg
4	Mass of an electron	9.11×10^{-31} kg
5	Specific charge	Charge ÷ mass (unit C kg^{-1})
6	Specific charge of a proton	9.58×10^{7} C kg^{-1}
7	Specific charge of an electron	1.76×10^{11} C kg^{-1}
8	Proton number (Z)	Number of protons
9	Nucleon number (A)	Number of protons + neutrons
10	Isotopes	Same number of protons, different number of neutrons

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AQA A-level Physics • Set 2: Stable and Unstable Nuclei **Paper 1**

#	Question	Answer
1	Role of the strong nuclear force	Holds the nucleus together, overcoming electrostatic repulsion between protons
2	Range over which the strong force is attractive	Up to about 3 fm
3	Strong force at separations below about 0.5 fm	Repulsive
4	Nuclei that undergo α decay	Very large nuclei (too many nucleons)
5	Nuclei that undergo β^- decay	Neutron-rich nuclei
6	What an α particle is	2 protons + 2 neutrons (a helium nucleus)
7	Change in *A* and *Z* in α decay	*A* decreases by 4, *Z* decreases by 2
8	Change in *A* and *Z* in β^- decay	*A* unchanged, *Z* increases by 1
9	Evidence that led to the neutrino being proposed	β particles are emitted with a range of energies
10	Why the neutrino was needed	To conserve energy (and momentum) in β decay

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AQA A-level Physics • Set 3: Particles, Antiparticles and Photons **Paper 1**

#	Question	Answer
1	Antiparticle of the electron	Positron
2	How an antiparticle compares with its particle	Same mass and rest energy, opposite charge
3	Rest energy of an electron	0.511 MeV
4	Rest energy of a proton	938 MeV
5	Photon energy equation	$E = hf = hc/\lambda$
6	Converting MeV to J	$\times 1.60 \times 10^{-13}$
7	Annihilation	Particle + antiparticle → two photons
8	Why annihilation produces two photons	To conserve momentum
9	Pair production	Photon → particle + antiparticle
10	Minimum photon energy for pair production	Total rest energy of the particle and antiparticle

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AQA A-level Physics • Set 4: Particle Interactions **Paper 1**

#	Question	Answer
1	Exchange particle for the electromagnetic force	Virtual photon
2	Exchange particles for the weak interaction	W^+ and W^- bosons
3	Exchange particle for the strong force between nucleons	Pion
4	Particles the strong interaction acts on	Hadrons only
5	Interaction responsible for β decay	Weak interaction
6	Exchange particle in β^- decay	W^-
7	Exchange particle in β^+ decay	W^+
8	Equation for β^- decay	$n \rightarrow p + e^- + \bar{\nu}_e$
9	Equation for β^+ decay	$p \rightarrow n + e^+ + \nu_e$
10	Equation for electron capture	$p + e^- \rightarrow n + \nu_e$

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AQA A-level Physics • Set 5: Classification of Particles Paper 1

#	Question	Answer
1	Hadrons	Particles that feel the strong interaction
2	Two classes of hadron	Baryons and mesons
3	Quark structure of a baryon	Three quarks
4	Quark structure of a meson	Quark + antiquark
5	The only stable baryon	Proton
6	Baryon number of a baryon, antibaryon and meson	+1, −1, 0
7	Products of free neutron decay	Proton, electron and antineutrino
8	Two types of meson in the AQA spec	Pions and kaons
9	Leptons	Particles that do not feel the strong interaction (e.g. electron, muon, neutrinos)
10	What a muon decays into	Electron (+ neutrinos)

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AQA A-level Physics • Set 6: Quarks and Antiquarks **Paper 1**

#	Question	Answer
1	Charge of an up quark	$+\frac{2}{3}e$
2	Charge of a down quark	$-\frac{1}{3}e$
3	Charge of a strange quark	$-\frac{1}{3}e$
4	Baryon number of a quark	$+\frac{1}{3}$
5	Strangeness of a strange quark	−1
6	Quark structure of a proton	uud
7	Quark structure of a neutron	udd
8	Quark structure of a π^+	$u\bar{d}$
9	Quark structure of a K^+	$u\bar{s}$
10	Quark change in β^- decay	$d \rightarrow u$

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AQA A-level Physics • Set 7: Conservation Laws **Paper 1**

#	Question	Answer
1	Quantities conserved in every particle interaction	Charge, baryon number, lepton number (plus energy and momentum)
2	Electron lepton number of an electron	$L_e = +1$
3	Electron lepton number of a positron	$L_e = -1$
4	Muon lepton number of a muon neutrino	$L_\mu = +1$
5	Baryon number of a lepton	0
6	Quantity conserved in strong but not weak interactions	Strangeness
7	Allowed change in strangeness in a weak interaction	0, +1 or −1
8	Why strange particles are produced in pairs	Made by the strong interaction, which conserves strangeness
9	Interaction by which strange particles decay	Weak
10	What kaons decay into	Pions (or leptons)

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AQA A-level Physics • Set 8: The Photoelectric Effect **Paper 1**

#	Question	Answer
1	Photoelectric effect	Electrons emitted from a metal surface when light above a threshold frequency shines on it
2	Threshold frequency	Minimum frequency of light that causes electrons to be emitted
3	Work function (φ)	Minimum energy needed to remove an electron from the metal surface
4	Effect of increasing intensity above the threshold frequency	More electrons emitted per second; $E_{k(max)}$ unchanged
5	Effect of increasing intensity below the threshold frequency	Still no electrons emitted
6	Effect of increasing frequency	$E_{k(max)}$ increases
7	Time delay before emission	None (emission is instant)
8	Why the wave theory cannot explain the effect	It predicts emission at any frequency if the light is intense enough
9	How the photon model explains the threshold frequency	One photon gives all its energy to one electron, so hf must be ≥ φ
10	Scientist who explained the effect using photons	Einstein

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AQA A-level Physics • Set 9: Photoelectric Equation and Stopping Potential Paper 1

#	Question	Answer
1	Photoelectric equation	$hf = \varphi + E_{k(max)}$
2	Threshold frequency equation	$f_0 = \varphi/h$
3	Why the kinetic energy is a maximum	Electrons below the surface lose energy escaping
4	Stopping potential (V_s)	Minimum p.d. needed to stop the fastest photoelectrons
5	Equation linking stopping potential and $E_{k(max)}$	$eV_s = E_{k(max)}$
6	Electronvolt (eV)	Energy gained by an electron accelerated through a p.d. of 1 V
7	1 eV in joules	1.60×10^{-19} J
8	Gradient of an $E_{k(max)}$ against f graph	h
9	x-intercept of an $E_{k(max)}$ against f graph	Threshold frequency f_0
10	y-intercept of an $E_{k(max)}$ against f graph	$-\varphi$

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AQA A-level Physics • Set 10: Energy Levels and Spectra **Paper 1**

#	Question	Answer
1	Excitation	An electron moves up to a higher energy level
2	Ionisation	An electron is removed from the atom
3	Ground state	Lowest energy level of an atom
4	Energy a photon needs to excite an atom	Exactly equal to the difference between two energy levels
5	Energy a colliding electron needs to excite an atom	Kinetic energy at least equal to the energy difference
6	Photon emitted when an electron moves down a level	$hf = E_1 - E_2$
7	How mercury atoms are excited in a fluorescent tube	Collisions with free electrons accelerated by the p.d.
8	Job of the phosphor coating	Absorbs UV photons and emits visible light
9	Line absorption spectrum	Dark lines on a continuous spectrum
10	What line spectra show	Electrons in atoms have discrete energy levels

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AQA A-level Physics • Set 11: Wave–Particle Duality **Paper 1**

#	Question	Answer
1	Evidence that light behaves as a wave	Diffraction and interference
2	Evidence that light behaves as a particle	Photoelectric effect
3	Evidence that electrons behave as waves	Electron diffraction
4	de Broglie equation	$\lambda = h/mv$
5	Quantity mv	Momentum
6	Effect of increasing electron speed on the diffraction pattern	Rings get closer together (λ decreases)
7	How electron speed is changed in an electron diffraction tube	Change the accelerating p.d.
8	Condition for significant diffraction	Wavelength similar to the gap or atomic spacing
9	Electron or proton at the same speed: longer λ	Electron (smaller mass)
10	Why a thrown ball shows no diffraction	Its wavelength is far too small